

Event counting and discontinuity of the experiential perfect¹

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Abstract. This paper investigates counting in the domain of events under the Mandarin experiential perfect marker *-guo*, with particular attention to the contrast between the two counting expressions, 「NUM *xia*」 and 「NUM *ci*」, for different levels of counting. The central question is why only certain activity verbs are compatible with 「NUM *xia*」 under *-guo*, and more generally what constrains the availability of event counting in this environment. I propose that the observed contrast is grounded in a distinction between two subclasses of activity verbs: those whose denotation is associated with determinate endpoints, and those whose denotation lacks such endpoints. Building on this distinction, I argue that the discontinuity effect of *-guo* discussed in the literature should be reformulated as a presupposition requiring the existence of determinate endpoints of the relevant events. Whether this presupposition is satisfied depends on the subclass of activities and the level of counting.

Keywords: event, countability, perfect, verbal classifiers.

1. Introduction

In Mandarin Chinese, the number of times an event occurs can be conveyed through the combination of a numeral and a verbal classifier, henceforth 「NUM CL」 (Li and Thompson, 1981). In (1a), for instance, *yi xia* ‘one time’ indicates that the hitting event occurred once; similarly, in (1b), *yi xia* suggests that the hair-combing event occurred once. In general, activity verbs are compatible with 「NUM CL」 (Huang, 1997; Zhang, 2017; Guo, 2021; Ye and Guo, 2023; a.o.).

- | | |
|-------------------------------------|---------------------------------------|
| (1) a. wo da-le ta yi xia. | b. ta shu-le yi xia toufa. |
| 1SG hit-PFV 3SG one CL _v | 3SG comb-PFV one CL _v hair |
| ‘I hit her one time.’ ² | ‘She combed hair one time.’ |

Within an event-counting expression 「NUM CL」, the numeral specifies the quantity of event occurrences, while the verbal classifier provides the counting unit and additional event-related information, including the level of counting. Observations across languages suggest that counting in the domain of events involves at least two hierarchically nested levels. These levels have been described under different terms in the literature (Andrews, 1983; Cinque, 1999; Larson, 2004; Landman, 2006; Deng, 2013; Donazzan, 2013; Bobaljik, 2017; Zhang, 2017; Ye and Guo, 2023; Guo, 2024; Wagiel, 2024; a.o.), beginning with Cusic’s (1981) distinction between events and occasions. A simplified illustration of levels of counting is through stacked *time*-adverbials in English, as shown in (2a). Sentence (2a) exhibits two distinct levels of counting (*three times twice*). Even though (2a) gives six individual knocks in total, it is not truth-conditionally equivalent to (2b). (2a) is true in a scenario where three consecutive knocks occur twice, as sketched by (3a), but false in a scenario in which there is only a single set of

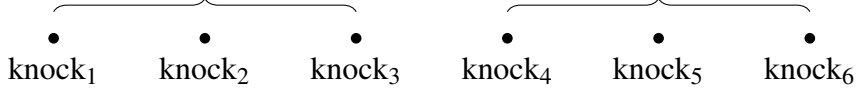
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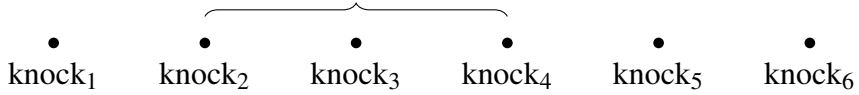
² The affix *-le* that follows the verb, also known in the literature as ‘the verbal *-le*’, is generally treated as a perfective morpheme. For simplicity, I follow Smith (1991), Lin (2005) among others and provide English translations of sentences marked by the perfective *-le* in the past tense.

three consecutive knocks alongside other individual knocks, as in (3b). By contrast, (2b) is true in both scenarios.

(2) a. John knocked on the door **three times twice**.

b. John knocked on the door **six times**.

(3) a.  (2a) ✓ (2b) ✓

b.  (2a) × (2b) ✓

These different levels of counting can also be realised through verbal classifiers in Mandarin Chinese (Deng, 2013; Donazzan, 2013; Zhang, 2017; Ye and Guo, 2023; Guo, 2024; Wagiel, 2024; a.o.). Among the vast collection of verbal classifiers, *xia* is used for counting at the lower level, hence glossed ‘CL_{vl}’, whereas *ci* is used for counting at the higher level, glossed ‘CL_{vh}’. For example, to express medical dosage instructions like ‘three sprays per application, twice daily’, *xia* is used to count the number of individual sprays, while *ci* is used to count the number of daily applications. Here, *xia* and *ci* occupy distinct positions and cannot switch positions, as demonstrated by (4). For now, I refer to these two levels of counting as lower-level and higher-level counting; the precise targets of *xia* and *ci* will be detailed in Section 3.

(4) a. wo meitian dei pen liang **ci** yao, mei **ci** pen san **xia**.

1SG everyday need spray two CL_{vh} medicine each CL_{vh} spray three CL_{vl}

‘Everyday I must spray the medicine two times, each time delivering three sprays.’

b.*wo meitian dei pen liang **xia** yao, mei **xia** pen san **ci**.

1SG everyday need spray two CL_{vl} medicine each CL_{vl} spray three CL_{vh}

Int.: ‘Everyday I must spray the medicine two times, each time delivering three sprays.’

Against this background, the interaction between event counting and aspect reveals an unexpected restriction, and this paper provides two new observations that help explain it. First, not all activity verbs can take event counting expressions under the experiential perfect marked by *-guo*, as shown by (5), even though the activity verb itself is compatible with *-guo*, as in (6). The contrast between (5a) and (5b) points to a distinction between two subclasses of activity verbs, which leads to a question: what distinguishes the two subclasses of activity verbs so that they behave differently with ‘NUM *xia*’ under *-guo*? (**Puzzle I**)

(5) a. wo da-**guo** ta **yi xia**.

1SG hit-EXP 3SG one CL_{vl}

‘I have hit him one time_(low).’³

b.*wo deng-**guo** ta **yi xia**.

1SG wait-EXP 3SG one CL_{vl}

Int.: ‘I have waited for him one time_(low).’

(6) a. wo da-**guo** ta.

1SG hit-EXP 3SG

‘I have hit him.’

b. wo deng-**guo** ta.

1SG wait-EXP 3SG

‘I have waited for him.’

Second, if *xia* for lower-level counting is replaced by *ci* for higher-level counting, all activity verbs can take ‘NUM *ci*’ under *-guo*, including the ones that do not work with ‘NUM *xia*’, as illustrated by (7). Comparing (7b) with (5b) gives rise to the second question: under what conditions does the experiential perfect *-guo* allow event counting? (**Puzzle II**)

³ To highlight different levels of counting in translation, where applicable, I use *time*-adverbials to translate higher-level counting (e.g. *two times*), and the combination of a numeral and a cognate object of the verb for lower-level counting (e.g., *three sprays*); see (4). Here, with verbs like *da* ‘hit’ and *deng* ‘wait’, cognate object constructions sound unnatural; I use subscripts *low* and *high* instead to mark counting at different levels explicitly in translation.

- (7) a. wo da-**guo** ta yi ci.
1SG hit-EXP 3SG one CL_{vh}
'I have hit him one time_(high).'
- b. wo deng-**guo** ta yi ci.
1SG wait-EXP 3SG one CL_{vh}
'I have waited for him one time_(high).'

To account for the distribution of activity verbs and $\lceil \text{NUM CL} \rceil$ under the experiential *-guo*, I propose a tripartite analysis. First, activity verbs can be subcategorised into Act_i (*da* 'hit') and Act_{ii} (*deng* 'wait'), depending on whether the events they denote are inherently discrete in the sense of having a **determinate endpoint** – in other words, whether they consist of stable atoms (cf. Chierchia, 2010). Second, what has been observed as the discontinuity effect of *-guo* is a presupposition regarding event properties. Essentially, *-guo* presupposes that the events in its input should have a unique maximal temporal gap that immediately succeeds the runtime of the relevant event, and that during this gap there are no such events. $\lceil \text{V}_{\text{Act}_i} \text{ NUM } xia \rceil$ satisfies this presupposition, as Act_i events have a determinate endpoint. In contrast, $\lceil \text{V}_{\text{Act}_{ii}} \text{ NUM } xia \rceil$ fails to meet this presupposition, as Act_{ii} events lack a determinate endpoint (**Puzzle I**). Third, placing the last occurrence of the Act_{ii} events inside a group of events coerces a determinate endpoint, which meets the presupposition of *-guo*, because groups are atomic (Landman, 1989a, b; Snyder and Shapiro, 2022; a.o.) and groups of events have determinate endpoints. This explains the well-formedness of $\lceil \text{V}_{\text{Act}_{ii}} \text{ NUM } ci \rceil$ under *-guo* (**Puzzle II**).

2. Activities and lower-level event counting

2.1. Availability of lower-level event counting under *-guo*

In Mandarin Chinese, *-guo* has been generally characterised as an experiential perfect marker (Li and Thompson, 1981; Smith, 1991; Klein et al., 2000; Lin, 2005; a.o.). A co-occurrence restriction is observed between *-guo* and predicates of different situation types within Vendler's (1957) classification. To start with, states like *shi* 'be' and *zhidao* 'know' are incompatible with *-guo*, as shown by (8) and (9).

- (8) a. ta shi xuesheng.
3SG be student
'She is a student.'
- b. *ta shi-**guo** xuesheng.
3SG be-EXP student
Intended: 'She has been a student.'
- (9) a. ta zhidao zhe ge guiding.
3SG know this CL_n rule
'She knows about this rule.'
- b. *ta zhidao-**guo** zhe ge guiding.
3SG know-EXP this CL_n rule
Intended: 'She has known about this rule.'

In the meanwhile, *-guo* can be found in sentences with achievements, accomplishments, and activities, as illustrated by (10)–(12) respectively.

- (10)a. ta dao-**guo** shanding.
3SG reach-EXP mountain.top
'She has reached mountain tops.'
- b. ta de-**guo** jinpai.
3SG receive-EXP golden.medal
'She has received a golden medal.'
- (11)a. ta da-sui-**guo** huaping.
3SG hit-break-EXP vase
'She has broken a vase.'
- b. ta jian-cheng-**guo** shuidianzhan.
3SG build-up-EXP hydropower.station
'She has built a hydropower station.'
- (12)a. ta da-**guo** wo.
3SG hit-EXP 1SG
'She has hit me.'
- b. ta shu-**guo** toufa.
3SG comb-EXP hair
'She has combed hair.'

Among the verbs compatible with *-guo*, only activity verbs can co-occur with lower-level counting expressions such as *yi xia* ‘one time_(low)’.⁴ They may appear in imperatives without an overt aspectual marker, as in (13)–(14), and in non-stative sentences with a perfective *-le*, as illustrated by (15)–(16).

- | | |
|---|---|
| <p>(13)a. ni da wo yi xia.
2SG hit 1SG one CL_{vl}
‘You hit me one time_(low).’</p> <p>(14)a. ni deng wo yi xia.
2SG wait 1SG one CL_{vl}
‘You wait for me one time_(low).’</p> <p>(15)a. ta da-le wo yi xia.
3SG hit-PFV 1SG one CL_{vl}
‘She hit me one time_(low).’</p> <p>(16)a. ta deng-le wo yi xia.
3SG wait-PFV 1SG one CL_{vl}
‘She waited for me one time_(low).’</p> | <p>b. ni shu yi xia toufa.
2SG comb one CL_{vl} hair
‘You comb hair one time_(low).’</p> <p>b. ni cha yi xia dizhi.
2SG check one CL_{vl} address
‘You check the address one time_(low).’</p> <p>b. ta shu-le yi xia toufa.
3SG comb-PFV one CL_{vl} hair
‘She combed hair one time_(low).’</p> <p>b. ta cha-le yi xia dizhi.
3SG check-PFV one CL_{vl} address
‘She checked the address one time_(low).’</p> |
|---|---|

Although activity verbs generally pattern alike in allowing *yi xia* ‘one time_(low)’ in imperatives and under the perfective *-le*, they do not behave uniformly under *-guo*. Some of the activity verbs modified by *yi xia* ‘one time_(low)’ are compatible with *-guo*, as shown in (17). Others, by contrast, cannot co-occur with *yi xia* ‘one time_(low)’ under *-guo*, as shown in (18).

- | | |
|---|---|
| <p>(17)a. ta meiyou da-guo wo yi xia.
3SG not hit-EXP 1SG one CL_{vl}
‘She has never once_(low) hit me.’</p> <p>(18)a.*ta meiyou deng-guo wo yi xia.
3SG not wait-EXP 1SG one CL_{vl}
Int.: ‘She has never once_(low) waited for me.’</p> | <p>b. ta zhi shu-guo yi xia toufa.
3SG only comb-EXP one CL_{vl} hair
‘She has only combed hair once_(low).’</p> <p>b.*ta zhi cha-guo yi xia dizhi.
3SG only check-EXP one CL_{vl} address
Int.: ‘She has only checked the address once_(low).’</p> |
|---|---|

2.2. Empirical motivations for subclassifying activities

The contrast between (17) and (18) in lower-level event counting under *-guo* is not between activities and other Vendlerian classes, but internal to activities, which calls for a finer-grained classification of activities. At first glance, this contrast appears to correlate with the distinction between activities in Vendler’s (1957) four-way classification and semelfactives (Smith, 1991; Rothstein, 2004). Semelfactive verbs denote a single occurrence of activities such as *knock* and *cough*, and are thus considered instantaneous. Similar to *da* ‘hit’, *qiao* ‘knock’ and *ke-sou* ‘cough’ can take lower-level event-counting expressions under *-guo*, as in (19). However, the contrast between (17) and (18) cannot be reduced to the contrast between activities and semelfactives, since activities compatible with lower-level event-counting under *-guo* are not unequivocally punctual. Alongside the examples in (17), there are a number of activity verbs that are clearly non-punctual, or can be interpreted non-punctually, but nevertheless remain compatible with *yi xia* ‘one time_(low)’ under *-guo*. Verbs of this kind include *yongbao* ‘hug’, *run* ‘wet’, *dong* ‘move’, *zou* ‘punch’, *za* ‘smash’, *yao* ‘shake’, *tui* ‘push’, *la* ‘pull’, *ca* ‘wipe’,

⁴ Ye and Guo (2023) provide a detailed account of why achievements reject *xia* for lower-level counting. Briefly, *xia* imposes a cumulativity presupposition of events and requires its input to contain summable events, while achievements are not summable at the lower level of counting.

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rou ‘rub’, *mo* ‘touch’, *ning* ‘screw’, *ji* ‘squeeze’, *an* ‘press’, and so on, as shown in (20).

- (19)a. ta meiyou **qiao-guo** yi **xia** men. b. ta zhi **kesou-guo** yi **xia**.
 3SG not knock-EXP one CL_{vl} door 3SG only cough-EXP one CL_{vl}
 ‘She has not given one knock on the door.’ ‘She has only given one cough.’
- (20)a. wo gen ta **yongbao-guo** yi **xia**. b. ta zhi **run-guo** yi **xia** zuichun.
 1SG and 3SG hug-EXP one CL_{vl} 3SG only wet-EXP one CL_{vl} lip
 ‘I have given him a hug.’ ‘She has only wetted lips one time_(low).’

The distinction between the two subclasses of activities, Activity I and Activity II (labelled Act_i and Act_{ii} for convenience), is also reflected in two further observations. First, Act_i and Act_{ii} differ in whether all numerals are equally acceptable with the verbal classifier *xia* for lower-level counting. Act_i can take any numeral that provides a precise counting result. For example, both *yi* ‘one’ and *san* ‘three’ can be the result of counting the occurrence of hitting events, as shown by (21). However, with Act_{ii} such as *deng* ‘wait’, only *yi* ‘one’ is allowed, yielding a vague counting result that merely indicates the existence of the event, as shown by (22).

- (21)a. ta **da-le** wo **yi** xia. b. ta **da-le** wo **san** xia.
 3SG hit-PFV 1SG one CL_{vl} 3SG hit-PFV 1SG three CL_{vl}
 ‘She hit me one time_(low).’ ‘She hit me three times_(low).’
- (22)a. ta **deng-le** wo **yi** xia. b.*ta **deng-le** wo **san** xia.
 3SG wait-PFV 1SG one CL_{vl} 3SG wait-PFV 1SG three CL_{vl}
 ‘She waited for me one time_(low).’ Int.: ‘She waited for me three times_(low).’

Second, Activity I and Activity II are not alike in their tolerance for temporal gaps. An Act_i event does not permit interruptions in any contexts. For instance, if a hitting event is interrupted by a brief pause, such as going to the bathroom, the hitting must be construed as two distinct events, as shown in (23a). By contrast, an Act_{ii} event can accommodate interruption. If someone who is waiting briefly pauses to take a bathroom break and then resumes waiting, what happens may still be construed as a single waiting event rather than two distinct waiting events, as illustrated in (23b).

- (23)a. wo da-le ta yi xia, qu cesuo hou you da-le ta yi xia.
 1SG hit-PFV 3SG one CL_{vl} go toilet after again hit-PFV 3SG one CL_{vl}
 ‘I hit her one time_(low). After taking a bathroom break, I hit her one more time_(low).’
 → Two hitting events
- b. wo deng-le ta yi xia, qu cesuo hou you deng-le ta yi xia.
 1SG wait-PFV 3SG one CL_{vl} go toilet after again wait-PFV 3SG one CL_{vl}
 ‘I waited for her one time_(low). After taking a bathroom break, I waited for her one more time_(low).’
 → One waiting event

2.3. Subclassifying activities via determinate endpoints

The two diagnostics just discussed, namely precise counting and tolerance for temporal interruptions, point towards a single underlying property of events: whether events exhibit stable atomicity (Ye and Guo, 2023), as illustrated in (24). This is a property developed by extending Chierchia’s (2010) notion of stable atomicity from the domain of individuals to the domain of events. Briefly, activities like hitting (Act_i) are stably atomic: an atomic hitting event stays atomic across all possible contexts. By contrast, there are also activities like waiting (Act_{ii}), for which a minimal waiting event may be identified across time as part of a larger waiting event.

(24) **Stable atomicity of events (Ye and Guo, 2023)**

$$\text{SAT}(P)(e) \stackrel{\text{def}}{=} \exists e' [e' \sqsubseteq e \wedge P(e') \wedge \forall c [\neg \exists e'' \text{ in } c [e'' \sqsubset e']]]$$

Building on the earlier notion of stable atomicity in the domain of events, I further propose that the crucial property underlying the contrast is the presence or absence of a clear, **determinate endpoint**. I treat the endpoint as the point at which a change of state occurs: prior to the endpoint, the event holds, whereas after the endpoint, no such event holds. Under this view, a determinate endpoint is a property of events which can be characterised in terms of the linear relation of **immediate succession** between intervals, together with a **gap**, i.e., an interval that starts right after the runtime of the event and contains no runtime of the same event. Basically, the event does not hold throughout this gap which immediately succeeds the runtime of the event. Then, for every P -event e with a determinate endpoint, there exists a maximal P -free interval that immediately succeeds the runtime of e . Under standard assumptions about linear time and temporal intervals, it follows that within a given context, the maximal immediately succeeding P -free interval is also unique. The determinate-endpoint property can be defined as follows, using basic notions from mereology and temporal semantics:

(25) **Determinate Endpoint (DEND)**

$$\text{DEND}(P) \stackrel{\text{def}}{=} \forall e [P(e) \rightarrow \exists t [\text{GAP}(t, e) \wedge \forall t' [\text{GAP}(t', e) \rightarrow t' \sqsubseteq t]]]$$

$$\text{where } \text{GAP}(t, e) \stackrel{\text{def}}{=} t \triangleright \tau(e) \wedge \neg \exists e' [P(e') \wedge \tau(e') \sqsubseteq t]$$

For a given P -event e , t is a P -free interval immediately succeeding $\tau(e)$,

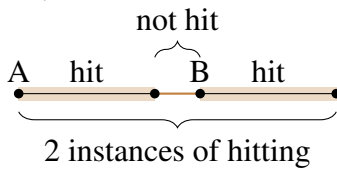
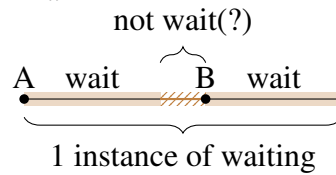
$$\text{and } t' \triangleright t \stackrel{\text{def}}{=} t \prec t' \wedge \neg \exists t'' [t'' \sqsubseteq t' \wedge t'' \sqsubseteq t] \wedge \neg \exists t''' [t \prec t''' \prec t']$$

t' immediately succeeds t .

Every P -event e has a unique maximal P -free interval immediately succeeding $\tau(e)$.

(26) presents a simplified illustration of the difference between the two subclasses of activities exemplified by *da* ‘hit’ and *deng* ‘wait’ respectively.

(26)

a. Act_i b. Act_{ii} 

An Act_i event, such as *da* ‘hit’, *shu* ‘comb’, *ti* ‘kick’, *dong* ‘move’, *ning* ‘screw’, *mo* ‘touch’, etc., has a determinate endpoint, and is thus inherently discrete. Taking *da* ‘hit’ as an example, each hitting event remains minimal regardless of contexts, and it has no subevent that also qualifies as a hitting event. In the meanwhile, the two adjacent hitting events are separated by a temporal gap during which no hitting takes place; see (26a). By contrast, an Act_{ii} event, such as *deng* ‘wait’, *pei* ‘accompany’, *ju* ‘hang out’, *kao* ‘test’, *cha* ‘check’, etc., lacks a determinate endpoint: in a given context, it may be unclear exactly when a waiting event comes to an end by looking at the process alone. Consequently, two temporally adjacent instances of waiting may be cross-temporally identified as a single event (cf. Landman, 2008; Lund, 2021), as illustrated in (26b). If the temporal gap between them is small or not cognitively salient, the two instances of waiting may be construed as one event, since a waiting event lacks an innate determinate endpoint. It is therefore unsurprising that Act_{ii} verbs such as *deng* ‘wait’ resist precise counting

by 「NUM *xia*」, as shown in (22): the existence of a determinate endpoint is a prerequisite for precise counting. By recasting stable atomicity as the presence of determinate endpoint, which is a more primitive property of events, the analysis does not have to make an ontological commitment to atoms.

2.4. Interim summary

Section 2 has shown the need to distinguish two subclasses of activities. In particular, compatibility with lower-level counting under the experiential perfect *-guo* correlates with whether an activity permits precise counting, and this pattern is also reflected in the tolerance for temporal gaps. Specifically, Activity I is the subclass that is compatible with lower-level counting under *-guo*. It includes predicates such as *da* ‘hit’, *bao* ‘hug’, *rou* ‘rub’, *ca* ‘wipe’, *dong* ‘move’. These activities permit precise lower-level counting with any numeral specifying the counting result, and a single instance of such events does not allow temporal gaps. By contrast, Activity II is incompatible with lower-level counting under *-guo*. It includes predicates such as *deng* ‘wait’, *cha* ‘check’, *ju* ‘get together’, *wan* ‘play’, *song* ‘see somebody off’. These activities resist precise lower-level counting, allowing only the numeral *yi* ‘one’ when the verbal classifier is *xia*, and they tolerate temporal gaps in the sense that two temporally adjacent instances may be construed as a single event in certain contexts. This subclassification of activities can be characterised in terms of the presence or absence of a determinate endpoint: Activity I has a determinate endpoint, whereas Activity II lacks one, as summarised in Table 1.

Table 1: Subclassification of activities

Subclass	Example	V- <i>guo</i> NUM CL _{vl}	V three CL _{vl}	Temporal gaps	Determinate endpoint
Activity I	<i>da</i> ‘hit’	+	+	–	+
Activity II	<i>deng</i> ‘wait’	–	–	+	–

Once the contrast between Activity I and Activity II has been characterised in terms of determinate endpoints, the next step is to ask how this property affects the acceptability of event counting under *-guo*. More specifically, the key is whether the relevant events in the sequence has a clear, well-defined determinate endpoint, which will be discussed in the next section.

3. Analysis

3.1. Discontinuity effect via determinate endpoint

Having identified an inherent property of events that distinguishes Activity I from Activity II, we can now ask why this property matters under the experiential perfect *-guo*. Taking the presence or absence of a determinate endpoint as the criterion for subclassifying activities explains the non-uniform behaviour of activity verbs with respect to precise lower-level counting and tolerance for temporal gaps. However, why must an event meet the criteria for precise counting in order for lower-level event counting to be acceptable in a *guo*-sentence? I argue that the answer lies in a requirement that *-guo* imposes on its input, namely, the denotation of 「V NUM *xia*」. This requirement has been widely discussed in the literature as the discontinuity effect of *guo*-sentences (Chao, 1968; Iljic, 1990; Smith, 1991; Lin, 2005, 2006; a.o.): *-guo* marks an experience that occurs prior to a reference time, and this experience no longer holds at the point

of reference. As shown in (27), the *guo*-sentence is incompatible with a subsequent clause that denotes a continuation of the crying event, in contrast to the corresponding *le*-sentence.

- (27)a. zuotian ta ku-guo, (#houlai ta yizhi ku).
 yesterday 3SG cry-EXP afterwards 3SG all.along cry
 ‘Yesterday, she had cried. (#Afterwards, she kept crying.)’
 b. zuotian ta ku-le, (houlai ta yizhi ku).
 yesterday 3SG cry-PFV afterwards 3SG all.along cry
 ‘Yesterday, she cried. (Afterwards, she kept crying.)’

On the basis of a fairly standard view of the asserted meaning of *-guo*, I propose that the discontinuity effect associated with *-guo* is in fact derived from a more primitive condition on events, namely the existence of a determinate endpoint, which is encoded as a presupposition of *-guo*. Before delving into that, let us first look at the asserted meaning of *-guo*. I follow the spirit of Lin (2005) and argue that the temporal meaning of *-guo* can be captured by the relation of three elements: Event Time, Topic Time, and Reference Time. Topic Time, based on Klein’s (1994) notion, is the time for which a sentence is asserted to be true, while Reference Time is the time of utterance. Essentially, *-guo* asserts that Event Time is located within Topic Time, and that Topic Time precedes Reference Time, which thereby yields the general past interpretation of *guo*-sentences. This can be illustrated independently of event counting. For example, in (28a), Event Time is the time when the publishing event happens, Topic Time is *qunian zhe ge shihou* ‘this time last year’, and Reference Time is by default the moment of speaking. Here, Event Time falls within the boundaries of Topic Time, but the acceptability of *-guo* depends on the temporal relation between Topic Time and Reference Time. As shown by (28), *-guo* is only acceptable when Topic Time precedes Reference Time. Under this characterisation, the temporal meaning of *-guo* relates a prior event to Reference Time, which is compatible with the anteriority theory of the perfect (Klein, 1994).

- (28)a. qunian zhe ge shihou, ta jiu yijing fabiao-guo lunwen le.
 last.year this CL time 3SG then already publish-EXP paper SFP
 ‘By this time last year, she had already published a paper.’
 b.*mingnian zhe ge shihou, ta jiu yijing fabiao-guo lunwen le.
 next.year this CL time 3SG then already publish-EXP paper SFP
 Int.: ‘By this time next year, she will already have published a paper.’

With the asserted meaning in place, I now turn to the other crucial component of the semantics of *-guo*: the determinate-endpoint presupposition. The lexical entry of *-guo* is given in (29). Specifically, I propose that *-guo* presupposes the existence of a determinate endpoint for the relevant event, which can be formalised as the DEND operator defined in (25), repeated here as (30): for every *P*-event, there exists a gap *t*, namely the maximal temporal interval that immediately succeeds the runtime of that event and contains no *P*-event. On this analysis, the determinate-endpoint presupposition picks out a unique maximal *P*-free interval immediately succeeding the runtime of the event, which means that the events compatible with *-guo* must be discrete, thereby deriving the discontinuity effect. This characterisation is also compatible with Pan and Lee’s (2004) representation of the ‘change-out-of-state’ property of *-guo* that requires a termination of the previous state.

(29) **Semantics of *-guo***

$\llbracket guo \rrbracket = \lambda P. \lambda t_{top}. \lambda t_{ref} : \mathbf{DEND}(P). \exists e [P(e) \wedge \tau(e) \sqsubseteq t_{top} \wedge t_{top} \prec t_{ref}]$
 (cf. Lin, 2005)

(30) **Determinate Endpoint Operator DEND**

$$\text{DEND}(P) \stackrel{\text{def}}{=} \forall e[P(e) \rightarrow \exists t[\text{GAP}(t, e) \wedge \forall t'[\text{GAP}(t', e) \rightarrow t' \sqsubseteq t]]]$$

$$\text{where } \text{GAP}(t, e) \stackrel{\text{def}}{=} t \triangleright \tau(e) \wedge \neg \exists e'[P(e') \wedge \tau(e') \sqsubseteq t]$$

For a given P -event e , t is a P -free interval immediately succeeding $\tau(e)$,

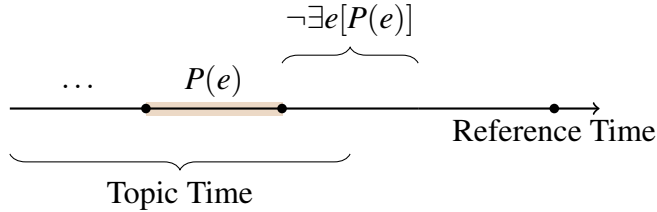
$$\text{and } t' \triangleright t \stackrel{\text{def}}{=} t \prec t' \wedge \neg \exists t''[t'' \sqsubseteq t' \wedge t'' \sqsubseteq t] \wedge \neg \exists t'''[t \prec t''' \prec t']$$

t' immediately succeeds t

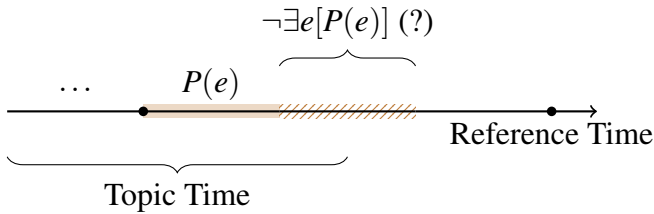
Every P -event e has a unique maximal P -free interval immediately succeeding $\tau(e)$.

(31) provides an interval-based temporal representation of *guo*-sentences in which a verb P is modified by a lower-level event-counting 「NUM *xia*」. Within a given context, for a *guo*-sentence of this kind, there are a number of P -events whose runtimes are located within Topic Time, and the relevant P -events share the same event-structural properties. There are two possible scenarios depending on whether the relevant P -events have a determinate endpoint, which can be viewed as a point of change from a P -interval to a P -free interval.

(31) a.



b.



In (31a), the P -event has a determinate endpoint. This guarantees the existence of a maximal P -free interval t that immediately succeeds the runtime of that event. This is the case with verbs of Activity I such as *da* ‘hit’, *shu* ‘comb’, *ti* ‘kick’, *mo* ‘touch’, and *rou* ‘rub’. As shown in Section 2, events denoted by these verbs have determinate endpoints. Therefore, the final occurrence in a sequence of Act_i events also has a determinate endpoint which guarantees the existence of an immediately succeeding maximal P -free gap. As a result, the presupposition of *-guo* is satisfied, and 「 $\forall_{\text{Act}_i}\text{-guo NUM } xia$ 」 is acceptable.

By contrast, in (31b), the relevant P -event lacks a determinate endpoint: by looking at the event itself, it is impossible to identify at which point the transition to a P -free interval occurs. As such, the existence of an immediately succeeding maximal P -free gap cannot be guaranteed. This is the case with verbs of Activity II (e.g., *deng* ‘wait’, *pei* ‘accompany’, *ju* ‘hang out together’, *cha* ‘check’, *chang* ‘taste’, ...). For some Act_{ii} events such as *deng* ‘wait’, not every relevant event can have a maximal gap, since two temporally adjacent instances of waiting can be cross-temporally identified as a single event, as shown by their tolerance for interruptions in (23b). For some Act_{ii} events like *chang* ‘taste’, none of the relevant events has an identifiable maximal gap, since the endpoint is not explicit by looking at the process of tasting alone. In

other words, what counts as completing the wine-tasting is not event-internally determined: the endpoint could be identified at swallowing, after another sip, or after the lingering flavour fades. Without a determinate endpoint, the presupposition of *-guo* is not met, which makes $\lceil V_{Act_{ii}}-guo \text{ NUM } xia \rceil$ is unacceptable.

The analysis developed so far explains why activity verbs pattern differently with lower-level event counting $\lceil \text{NUM } xia \rceil$ under *-guo* (**Puzzle I**). In contrast to verbs of Activity I (e.g., *da* ‘hit’), verbs of Activity II (e.g., *deng* ‘wait’) cannot be modified by $\lceil \text{NUM } xia \rceil$ in a *guo*-sentence ($*V_{Act_{ii}}-guo \text{ NUM } xia$), and they are incompatible with precise counting at the lower level ($*V_{Act_{ii}}$ three *xia*). However, once the verbal classifier *xia* is replaced by *ci*, the contrast between the two subclasses disappears: verbs of Activity II are compatible with $\lceil \text{NUM } ci \rceil$ under *-guo*, just like verbs of Activity I (**Puzzle II**), as shown earlier in (7). Why does a change in the level of counting affect the compatibility of verbs of Activity II with event counting in *guo*-sentences? In the next subsection, I argue that higher-level counting introduces a clear boundary, which then coerces a determinate endpoint event-externally that satisfies the presupposition of *-guo*.

3.2. A way to coerce an endpoint

The remaining question is why the same Activity II predicates become acceptable once the level of counting changes to the higher level. As detailed in Section 2, Activity II events (e.g., *deng* ‘wait’) cannot be precisely counted at the lower level: they reject most numerals when co-occurring with $\lceil \text{NUM } xia \rceil$ except for *yi* ‘one’, as shown by (32a). However, this restriction on numerals does not apply for higher-level counting. Verbs of Activity II such as *deng* ‘wait’ are compatible with any numeral in $\lceil \text{NUM } ci \rceil$, as shown by (32b).

- (32)a. wo deng-le ta {**yi** / ***liang** / ***san** / ***shi**} **xia**.
 1SG wait-PFV 3SG one two three ten CL_{vl}
 ‘I waited for him {one / #two / #three / #ten} times_(low).’
- b. wo deng-le ta {**yi** / **liang** / **san** / **shi**} **ci**.
 1SG wait-PFV 3SG one two three ten CL_{vh}
 ‘I waited for him {one / two / three / ten} times_(high).’

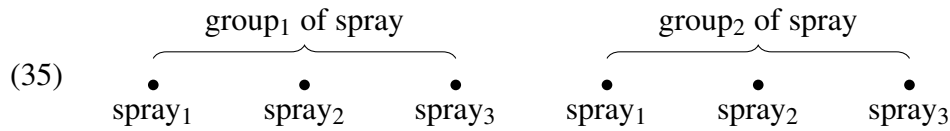
A comparable change in acceptability appears in *guo*-sentences containing verbs of Activity II. In contrast to lower-level counting, higher-level counting is available with verbs of Activity II under *-guo*, as shown in (33).

- (33)a. wo deng-guo ta {***yi** / ***liang** / ***san** / ***shi**} **xia**.
 1SG wait-EXP 3SG one two three ten CL_{vl}
 ‘I have waited for him {#one / #two / #three / #ten} times_(low).’
- b. wo deng-guo ta {**yi** / **liang** / **san** / **shi**} **ci**.
 1SG wait-EXP 3SG one two three ten CL_{vh}
 ‘I have waited for him {one / two / three / ten} times_(high).’

For a sequence of events lacking determinate endpoints like *deng* ‘wait’ in (32) and (33), it is impossible to identify a unique maximal *P*-free gap that immediately succeeds the runtime of the relevant *P*-event, and the presupposition of *-guo* is not satisfied. However, even if the event itself lacks a determinate endpoint, a determinate endpoint can be coerced, and higher-level counting provides a way of doing so. As introduced in Section 1, the verbal classifier *ci* provides a unit for higher-level counting in the domain of events (Deng, 2013; Donazzan, 2013;

Zhang, 2017; Ye and Guo, 2023; Guo, 2024; Wagiel, 2024), which intuitively corresponds to the **occasion** level in Cusic’s (1981) system. In the medicine-applying situation described by (4), repeated here as (34), the target counted by *ci* can be subdivided into the target of *xia*: one daily application of medicine is composed of several individual sprays of medicine, as shown by (34a). In the meanwhile, the target of *xia* cannot be subdivided into the target of *ci*; see (34b). On the basis of this relation, I adopt the notion of **group** (Link, 1984; Landman, 1989a, b) and characterise the target of *ci* as **groups of events**, while the target of *xia* is **events**. That is to say, the sentence in (34a) describes two groups of medicine-spraying events, each group consisting of three medicine-spraying events, as illustrated by (35).

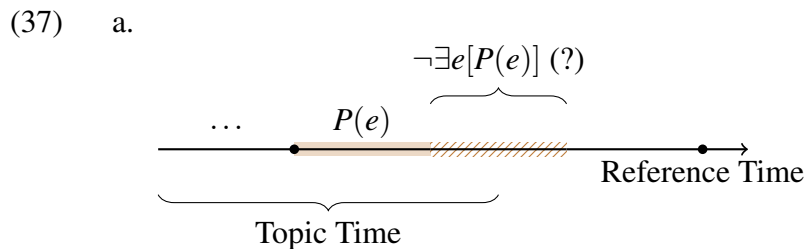
- (34)a. wo meitian dei pen liang **ci** yao, mei **ci** pen san **xia**.
 1SG everyday need spray two CL_{vh} medicine each CL_{vh} spray three CL_{vl}
 ‘Everyday I must spray the medicine two times, each time delivering three sprays.’
 b. *wo meitian dei pen liang **xia** yao, mei **xia** pen san **ci**.
 1SG everyday need spray two CL_{vl} medicine each CL_{vl} spray three CL_{vh}
 Int.: ‘Everyday I must spray the medicine two times, each time delivering three sprays.’

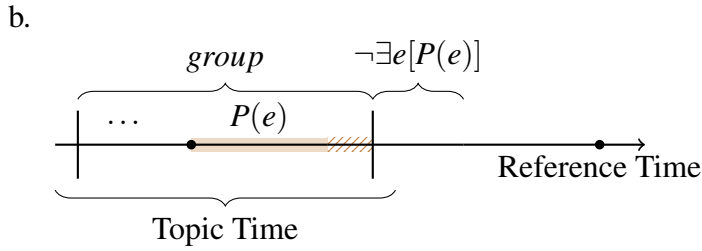


Following general assumptions in Neo-Davidsonian event semantics, the denotations of *pen san xia* ‘spray three CL_{vl}’ and *pen liang ci* ‘spray two CL_{vh}’ in (34a) can be represented as (36), where the group forming operator $\uparrow$ (Link, 1984; Landman, 1989a, b) is used to characterise groups of events.

- (36)a. $\llbracket \text{spray three } xia \rrbracket = \{e_{\text{spray}1} \oplus e_{\text{spray}2} \oplus e_{\text{spray}3}, \dots\}$
 b. $\llbracket \text{spray two } ci \rrbracket = \{\uparrow(e_{\text{spray}1} \oplus e_{\text{spray}2} \oplus e_{\text{spray}3}) \oplus \uparrow(e_{\text{spray}1} \oplus e_{\text{spray}2} \oplus e_{\text{spray}3}), \dots\}$

Given that groups are atomic (Link, 1984; Landman, 1989a, 2000, 2006; Snyder and Shapiro, 2022), they can be precisely counted. Groups of events are thus expected to have clear-cut boundaries, and hence a determinate endpoint. Since a group supplies a determinate endpoint for the events it contains, groups of Act_{ii} events are compatible with precise counting, as demonstrated in (32b). Therefore, despite the fact that Act_{ii} events lack an inherent determinate endpoint, as in (37a), a determinate endpoint can be coerced by placing the final occurrence of such *P*-events within a group of *P*-events, while ensuring that Reference Time remains outside that group, or if it makes more sense, that occasion. In this way, the existence of a *P*-free interval can be guaranteed, as illustrated in (37b). Through the determinate endpoint supplied by groups, the presupposition of *-guo* is satisfied, and verbs of Activity II modified by $\lceil \text{NUM } ci \rceil$ are acceptable under *-guo*.





In addition, verbs of Activity II can occur under *-guo* on their own, despite the absence of an inherent determinate endpoint, as shown in (38). This is also because a determinate endpoint can be coerced. A single event is by default interpreted at the occasion level where external components may supply the relevant endpoint. For example, in (38a), the endpoint of waiting can be coerced by whether the person waited for has arrived, or by the agent's intention. In (38b), the endpoint of hanging out together can be coerced by the overall spatial location of the participants or by the arrival or departure of some salient participants. With a coerced determinate endpoint, there must be a maximal *P*-free gap that immediately succeeds the runtime of the occasion and satisfies the presupposition of *-guo*.

(38)a. wo deng-**guo** ta.
1SG wait-EXP 3SG
'I have waited for him.'

b. women ju-**guo**.
1SG hang.out.together-EXP
'We have hung out together.'

3.3. Prediction

This analysis characterises the discontinuity effect observed in sentences marked by the experiential perfect *-guo* as a presupposition requiring the existence of a determinate endpoint for its input, whether this input is an event or a group of events. In lower-level (event-level) counting under *-guo*, this requirement gives rise to a contrast in acceptability between verbs of Activity I and Activity II. The perfective marker *-le*, on the other hand, does not impose such a requirement on its input. As illustrated in (39), with all other elements being the same, the *guo*-sentence entails that the leg has recovered from the breaking event by Reference Time, while the corresponding *le*-sentence allows an interpretation where the leg might still be broken at Reference Time.

(39)a. ta de tui shuai-duan-**guo**.
3SG MOD leg fall-break-EXP
'His leg had been broken.'

→ no longer broken

b. ta de tui shuai-duan-**le**.
3SG MOD leg fall-break-PFV
'His leg was broken.'

→ still broken

Given the difference between *-guo* and *-le*, this analysis correctly predicts that the acceptability contrast between verbs of Activity I (e.g., *da* 'hit') and Activity II (e.g., *deng* 'wait') with event-level counting under *-guo* is not observed in sentences marked by *-le*: both subclasses of activity verbs are compatible with *yi xia* 'once_{low}' under *-le*, as shown in (40).

(40)a. wo da-le ta yi xia.
1SG hit-PFV 3SG one CL_{vl}
'I hit him once_(low).'

b. wo deng-le ta yi xia.
1SG wait-PFV 3SG one CL_{vl}
'I waited for him once_(low).'

4. Conclusion

This work examines the compatibility of activity verbs and two types of event-counting expressions ($\lceil \text{NUM } xia \rceil$ and $\lceil \text{NUM } ci \rceil$) under the experiential perfect marker *-guo* in Mandarin Chinese. It addresses two puzzles: first, why activity verbs differ in their compatibility with $\lceil \text{NUM } xia \rceil$ under *-guo* (**Puzzle I**); and second, under what conditions event counting is available under *-guo* (**Puzzle II**).

The preceding sections derive both puzzles from a single interaction between the level of counting and the determinate-endpoint presupposition of *-guo*. First, activity verbs can be divided into two subclasses, Activity I (e.g., *da* ‘hit’) and Activity II (e.g., *deng* ‘wait’), depending on whether the events they denote have a determinate endpoint.

Second, the discontinuity effect of the experiential perfect marker *-guo* can be derived from the determinate-endpoint presupposition. More specifically, in addition to asserting that Event Time is located within Topic Time, and that Topic Time precedes Reference Time, *-guo* presupposes the existence of a maximal interval that immediately succeeds the runtime of that event, during which there are no relevant event. Events of Activity I, such as *da* ‘hit’, have determinate endpoints and therefore guarantee the existence of such an interval, which satisfies the presupposition of *-guo*. As a result, $\lceil V_{Act_i} - guo \text{ NUM } xia \rceil$ is acceptable. By contrast, events of Activity II, such as *deng* ‘wait’, lack determinate endpoints. Consequently, not every event can be immediately succeeded by a maximal interval during which there is no such event. At the event level, Activity II fails to meet the determinate-endpoint presupposition of *-guo*, and $\lceil V_{Act_{ii}} - guo \text{ NUM } xia \rceil$ is unacceptable. This resolves **Puzzle I**.

Third, at the level of groups of events, the presupposition of *-guo* can be satisfied by all activity verbs. Groups of events are atomic in nature and thus inherently associated with determinate endpoints, and thus they can coerce a determinate endpoint and thereby satisfy the presupposition of *-guo*. This resolves **Puzzle II**.

A prediction of this analysis is that, with the Mandarin perfective marker *-le* which does not exhibit a comparable discontinuity effect, the two subclasses of activities should not differ in their compatibility with *yi xia* ‘once_{low}’ for counting at the event level. This prediction is indeed borne out in Mandarin.

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